

Solutions

Question 7

(a) $\vec{r}(t) = \langle \sqrt{t-1}, \frac{t}{2} \rangle$ (x -component = $\sqrt{t-1}$, y -component = $\frac{t}{2}$)

Eliminate the parameter: I want to have an equation in x and y and relate them both. They are both time-dependent, but maybe I don't care about time!

$$y = \frac{t}{2} \quad \leftarrow \text{need to sub in } x$$

$$x = \sqrt{t-1} \quad \longrightarrow \quad x^2 + 1 = t$$

$$\boxed{y = \frac{1}{2}(x^2 + 1)}$$

(b) $y = \frac{1}{2}(x^2 + 1)$ (a parabola)

$$y(0) = \frac{1}{2}(0^2 + 1) = \frac{1}{2}$$

- Vertex at the y -intercept.

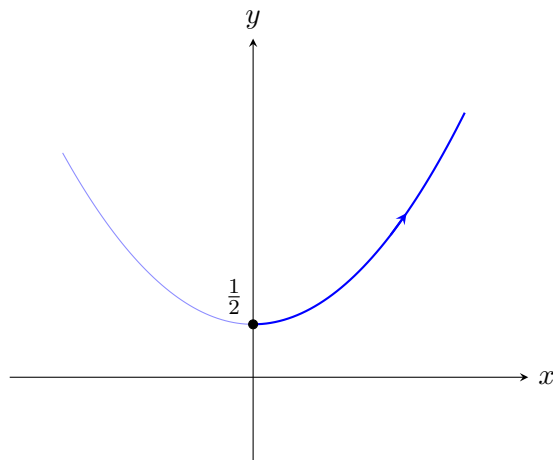
Look at the equations $x = \sqrt{t-1}$ and $y = t/2$.

- t can be any value for $y = t/2$, but $t \geq 1$ must be true for $x = \sqrt{t-1}$.

Minimum $t = 1$. At $t = 1$:

$$x = \sqrt{1-1} = \sqrt{0} = 0, \quad y = \frac{1}{2}.$$

As t increases, x and y increase, so the arrow is in the up direction.



$$y = \frac{1}{2} (x^2 + 1), \quad \vec{r}(t) = \langle \sqrt{t-1}, \frac{t}{2} \rangle$$

Question 8

(a)

$$\begin{aligned} \vec{r}_1(t) &= \langle \underbrace{t-1}_{x_1}, \underbrace{3t-t^2}_{y_1} \rangle \\ \vec{r}_2(t) &= \langle \underbrace{-t-1}_{x_2}, \underbrace{\frac{1}{4}t^2+2t}_{y_2} \rangle \end{aligned}$$

Collision: $\vec{r}_1$ and $\vec{r}_2$ equal each other (i.e. hit the same place) at the *same* time t .

So let's equate components and solve for time! This gives two equations:

$$\begin{aligned} t-1 &= -t-1 \\ 3t-t^2 &= \frac{1}{4}t^2+2t \end{aligned}$$

From the first equation:

$$t-1 = -t-1 \Rightarrow 2t = 0 \Rightarrow t = 0$$

Check the other equation:

$$3(0) - (0)^2 = \frac{1}{4}(0)^2 + 2(0) \quad \checkmark \quad \text{Yes, } t = 0 \text{ works.}$$

Sub in $t = 0$ to either $\vec{r}_1$ or $\vec{r}_2$:

$$\begin{aligned} x_1 &= x_2 = 0-1 = -1 \\ y_1 &= y_2 = 3(0) - 0^2 = 0 \end{aligned}$$

Therefore $\langle -1, 0 \rangle$ is the collision location.

$$\therefore \text{Particles collide at } t = 0. \text{ Collision location} = \langle -1, 0 \rangle.$$

(b)

$$\begin{aligned} \vec{r}_1(t) &= \langle \underbrace{t-1}_{x_1}, \underbrace{3t-t^2}_{y_1} \rangle \\ \vec{r}_2(t) &= \langle \underbrace{-t-1}_{x_2}, \underbrace{\frac{1}{4}t^2+2t}_{y_2} \rangle \end{aligned}$$

Intersection: $\vec{r}_1$ and $\vec{r}_2$ will equal each other (hit the same place) at *different* times.

Let's say $\vec{r}_1$ hits that place at time t and $\vec{r}_2$ hits that place at time s .

Equate the x and y components. This gives two equations:

$$\begin{aligned}t - 1 &= -s - 1 \\3t - t^2 &= \frac{1}{4}s^2 + 2s\end{aligned}$$

From the first equation, $t = -s$, i.e. $s = -t$.

Substitute $s = -t$ into the second equation:

$$\begin{aligned}3t - t^2 &= \frac{1}{4}(-t)^2 + 2(-t) \\3t - t^2 &= \frac{1}{4}t^2 - 2t \\ \Rightarrow -\frac{5}{4}t^2 + 5t &= 0\end{aligned}$$

- Solve (e.g. with the quadratic formula or by factoring): $t = 0$ and $t = 4$, so $s = -4$.
- $t = 0$ is the collision solution.
- Find the intersection location:

$$\vec{r}_1(t) = \vec{r}_1(4) = \langle 4 - 1, 3(4) - 4^2 \rangle = \langle 3, -4 \rangle$$

Sanity check with $s = -4$ for $\vec{r}_2$:

$$\vec{r}_2(-4) = \langle -(-4) - 1, \frac{1}{4}(-4)^2 + 2(-4) \rangle = \langle 3, -4 \rangle \quad \checkmark$$

$\therefore$ The particles intersect. They intersect at $(3, -4)$. $\vec{r}_1$ intersects at $t = 4$ and $\vec{r}_2$ intersects at $t = -4$.

(Negative time is allowed in 171!)

Question 9

Consider the curve defined by the equation

$$(x - 3)^2 + (y - 7)^2 = 36 \tag{*}$$

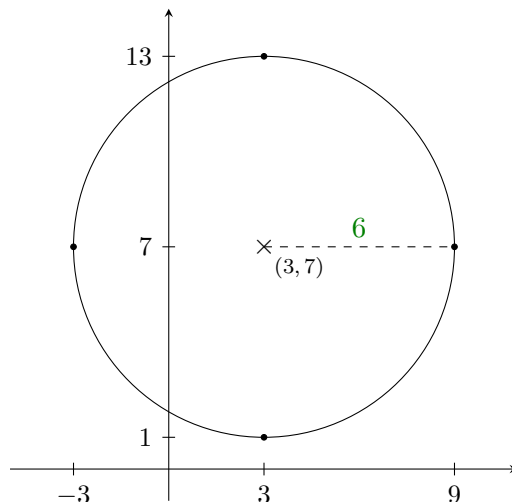
(a) Sketch this curve.

Recall: The equation for a circle centered at (a, b) with radius r is given by

$$(x - a)^2 + (y - b)^2 = r^2.$$

Thus, looking at equation $(\star)$ we have:

$$(x - 3)^2 + (y - 7)^2 = 6^2 \quad (= 36)$$



(b) Find a vector-valued function $\mathbf{u}(t)$ that would follow that curve over time.

$$\mathbf{u}(t) = \langle 6 \cos(t) + 3, 6 \sin(t) + 7 \rangle.$$

(Intuition for this solution is provided in the following question (Question 10))

(c) Show by a calculation that your $\mathbf{u}(t)$ always lies on the given curve.

Note: $\mathbf{u}(t) = \langle x(t), y(t) \rangle = \langle 6 \cos(t) + 3, 6 \sin(t) + 7 \rangle$.

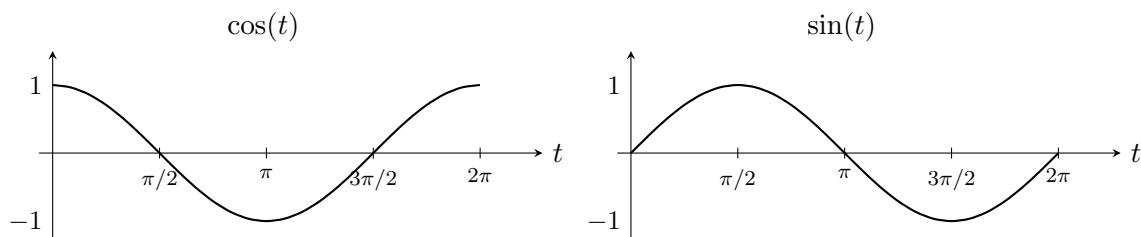
Plugging $x(t)$ and $y(t)$ into $(\star)$ gives us:

$$\begin{aligned} (\star) &= (x - 3)^2 + (y - 7)^2 = ((6 \cos(t) + 3) - 3)^2 + ((6 \sin(t) + 7) - 7)^2 \\ &= (6 \cos(t))^2 + (6 \sin(t))^2 \\ &= 6^2 (\cos^2(t) + \sin^2(t)) \\ &= 6^2 (1) = 36. \end{aligned}$$

Question 10

For each of the circles shown below, write down a parametric function that would trace out that trajectory, and match any additional properties given. Multiple answers are possible.

Recall:



Important to remember at $t = 0$:

- $\cos(t)$ starts at a maximum
- $-\cos(t)$ starts at a minimum
- $\sin(t)$ starts at 0 and increases initially
- $-\sin(t)$ starts at 0 and decreases initially.

You can scale each trig function by the radius of your circle, i.e. $\pm r \cos(t)$, $\pm r \sin(t)$.

(a) Starts at $(3, 0)$. That is, $x(0)$ is at a maximum, and $y(0)$ is at 0 and increasing initially.

This tells you we will be using $3 \cos(t)$ for $x(t)$ and $3 \sin(t)$ for $y(t)$.

We are centered at the origin, so no shift is required. Thus,

$$\vec{C}_1(t) = \langle 3 \cos(t), 3 \sin(t) \rangle.$$

(b) Similar argument as (a), except we are now centered at $(3, -2)$ with $r = 2$. Thus,

$$\vec{C}_2(t) = \langle 2 \cos(t) + 3, 2 \sin(t) - 2 \rangle.$$

(c) Starts at $x(0) = 0$ with $x(t)$ decreasing initially, and $y(0) = -1$, which is a minimum.

Thus $x(t) = -\sin(t)$ and $y(t) = -\cos(t)$. Thus,

$$\vec{C}_3(t) = \langle -\sin(t), -\cos(t) \rangle.$$